Amit Brilant

Data Scientist · Computer Vision & Medical Imaging

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Data scientist specializing in machine learning, deep learning, computer vision and image processing, with nearly 4 years as a Data & AI Analyst at a medical device company. Extensive hands-on experience analyzing clinical trial data and developing internal AI tools with LLMs. M.Sc. in Electrical and Computer Engineering, with thesis research in medical imaging (paper under journal review), and B.Sc. in Medical Engineering.

SKILLS

Languages: Python, SQL, MATLAB

ML & Computer Vision: PyTorch, scikit-learn, XGBoost, CNNs, U-Net, YOLO, Segment Anything (SAM), self-supervised learning, image processing, OpenCV, Grad-CAM, SHAP

LLMs & AI Applications: OpenAI & Anthropic APIs, RAG, AI agents, tool calling, evals, Hugging Face

Data & Tools: pandas, NumPy, SciPy, statistics, EDA, ETL pipelines, Plotly, Git, Jupyter, Streamlit, Claude Code

EXPERIENCE

EyeYon Medical · ophthalmic medical devices — Jan 2023 to Present

Data & AI Analyst · *Part-time*

- Clean, validate and analyze complex clinical trial data from EDC systems, detecting errors and anomalies, and deliver EDA, statistical analyses, recurring reports and dashboards used in ongoing trial management.
- Surfaced patterns in clinical outcomes through in-depth statistical analyses, prompting internal investigations into patient management and the device's biological effects.
- Built Python pipelines that extract data from multiple EDC sources, and consolidated patient, surgeon, clinical trial and commercial data into one structured data layer, presenting insights to management, R&D, medical, product and commercial teams to support data-driven decisions.
- Mapped bottlenecks across departments and built internal AI tools with LLM APIs: a publications portal shared with physicians, automated monitoring of news, literature, tenders and regulatory updates, and a RAG chatbot prototype over company knowledge.
- Led company-wide training on ChatGPT, Claude and building AI agents.

Graduate Research Student, Computer Vision & Deep Learning · Ben-Gurion University of the Negev — Jun 2024 to Sep 2026

- Led an M.Sc. research project in collaboration with Meir Medical Center: designed, trained and evaluated deep-learning models for ovarian carcinoma diagnosis from ultrasound.

Data Collection Assistant · Binah.ai · *Part-time* — Dec 2021 to Oct 2022

- Collected vital-sign data with gold-standard devices and ran validation and accuracy tests for a video-based health app.

RESEARCH & PROJECTS

Three-Stage Computer-Vision Pipeline for Ovarian Carcinoma Diagnosis from Ultrasound — 2026

M.Sc. thesis, with Meir Medical Center · *github.com/Amit-Brilant/ovarian-cancer-tvus-pipeline*

EfficientNet-B7 triage, YOLOv8m lesion localisation, and benign-versus-malignant classification fusing CNN embeddings with 11 shape descriptors from Segment Anything masks: 0.952 accuracy, 0.964 AUC (patient level). Grad-CAM and SHAP. Released the de-identified dataset (1,082 images, 260 patients) and weights.

Self-Supervised Cell Segmentation with BYOL and U-Net — 2024

M.Sc. course project, Ben-Gurion University · *github.com/Amit-Brilant/cell-segmentation-byol-unet*

BYOL pretraining of a U-Net encoder on unlabelled microscopy images, then supervised fine-tuning for 3-class segmentation. Mean SEG improved by up to 12.4% over the supervised baseline, with the largest gains at the smallest labelled sets.

Virtual Biopsy for Early Melanoma Detection — 2022

B.Sc. final project, with Sheba Medical Center · *github.com/Amit-Brilant/rcm-nevi-composite-pattern-classification*

ConvNeXt classification of reflectance confocal microscopy tiles with composite cellular patterns, using a clinically weighted penalty matrix built into both the loss function and the evaluation metric. 90.7% clinically weighted accuracy on 717 held-out tiles.

EDUCATION

M.Sc. Electrical and Computer Engineering · Ben-Gurion University of the Negev — Dec 2023 to Sep 2026

Thesis: *Detection and Diagnosis of Ovarian Carcinoma from Complex Cysts Using Computer Vision and Ultrasound Imaging.*

Paper under journal review.

B.Sc. Medical Engineering · Afeka Tel Aviv Academic College of Engineering — Oct 2018 to Jul 2022

Major: Data Science and Medical Information Systems.

AfekaVision · Student computer-vision group: weekly Kaggle competitions and paper reviews (Nov 2021 to Jul 2022).

Certifications: DeepLearning.AI: Deep Learning, Machine Learning, Probability & Statistics · U. Michigan: Deep Learning for Computer Vision